Random-Order Prefix Assembly of a Frozen Hénon Core

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Abstract

We study the first prefix of a uniformly random ordering of the sixteen named labels whose generated subgroup is the full finite core. Exhaustive closure enumeration and a pivotal-label counting identity give the exact distribution over all $16! = 20922789888000$ permutations. The stopping time ranges from 3 to 16, with exact mean $36499/3960 \approx 9.2169$. An independent closure checker, a symbolic factorial generating check, clean replay, and fifteen hostile mutations certify the receipt. This is a finite named-support result under the firewall `NO_BAD_EULER_OR_ROOT_NUMBER`.

1 Stopping rule

Let π be a permutation of $L = \{S_1, \dots, S_{16}\}$ and $A_k(\pi) = \{\pi_1, \dots, \pi_k\}$. Define

$$T(\pi) = \min\{k : \Phi(A_k(\pi)) = Q\}.$$

The C75 point-set closure gives 25 inclusion-minimal full-core supports, all of size three, but larger prefixes can be the first full prefix when their last label is pivotal.

Theorem 1 (Pivotal-prefix identity). *For a full-core support S of size k, let*

$$p(S) = \#\{\ell \in S : \Phi(S \setminus \{\ell\}) \neq Q\}.$$

Then

$$N_k = \#\{\pi : T(\pi) = k\} = \sum_{|S|=k,\, \Phi(S)=Q} p(S)(k-1)!(16-k)!.$$

Proof. Choose the first k labels to be S, choose a pivotal label ℓ to be last within that prefix, order the other $k-1$ labels arbitrarily, and order the remaining $16-k$ labels arbitrarily. The preceding prefix is not full, and every permutation with stopping time k is counted exactly once. $\square$

2 Exact distribution

The stopping counts for $k = 3, \dots, 16$ are

k	N_k
3	934053120000
4	1641059481600
5	1927502438400
6	1927328256000
7	1807490764800
8	1671222067200
9	1556813260800
10	1467573811200
11	1398684672000
12	1348868505600
13	1319170406400
14	1307674368000
15	1307674368000
16	1307674368000

They sum to 16!. The probability generating function $G(z) = \sum_k N_k z^k/16!$ satisfies $G(1) = 1$ and

$$G'(1) = \mathbb{E}[T] = \frac{36499}{3960}.$$

For example, $\Pr[T = 3] = 5/112$ and $\Pr[T \geq 15] = 3/16$.

3 Audit and scope

The producer binds C76/C78 and the frozen C81 receipt, reconstructs the C75 closure table, and computes every full-support and pivotal pattern. The independent checker repeats this route and verifies all factorial-weighted counts, reduced probabilities, survival counts, and the expectation. A SymPy polynomial check independently recovers the stopping counts. The canonical evidence hash is `4777695a3082a2cca1ee82cdced208f0bddf56431285774a51e7563c4cfdfea0`. No arithmetic/local, Euler-factor, root-number, automorphy, full Burnside/table-of-marks, or Hilbert–Polya conclusion is asserted.